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Swaptions from a Clearinghouse perspective

**Hedging swaptions, an option on
interest rate swaps, using
compression.**

Joel Forsberg

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Swaptions from a Clearinghouse perspective.
Hedging swaptions, an option on interest rate swaps, using compression from a clearinghouse perspective.

Department of physics
Umeå University
Umeå, Sweden

Examiner:
Lisa Hed, Umeå University

Supervisors:
Markus Nyberg, Nasdaq
Jonas Segervald, Umeå University.

Abstract

With the increasing popularity of interest rate swaps the need to understand swaptions, an option of an interest rate swap, is of great importance. A swaption can be used in both speculative purposes and to hedge against changes in interest rates. The most important thing to understand is the pricing formula. By starting at the basic rate instrument, bonds, we will work our way towards the pricing formula for a swaption, the Black76 model. The Black76 model is a variant of the Nobel prize winning formula Black-Scholes-Merton. With the pricing model we can start looking at the main scope of this thesis, a hedging strategy against swaptions from a clearinghouse perspective.

Clearinghouses are central to the modern financial market. They act as a middleman in order to clear trades from clearing members and have an oversight of the financial market. In case a clearing member defaults, the clearinghouse will gain control over the defaulted portfolio. The clearinghouse will host an auction of the portfolio which they strive to hold after 5 to 15 days. When they hold the portfolio, they are exposed to the risks and therefore it's of great importance to be able to hedge the assets in the portfolio. In this thesis a strategy and algorithm have been developed to delta-hedge swaptions in order to be delta-neutral under stable market conditions.

In the thesis we will consider two cases. The first case is when the clearinghouse receives the portfolio long before the swaptions maturity. In this case forward swaps are used to hedge and in order to reduce the number of forward swaps obtained, compression is used. The second case is when the swaption maturity will be reached within the period the clearinghouse holds the portfolio. For the days before maturity is reached, forward swaps and compression is used. After maturity is reached interest rate swaps is used to hedge.

For both cases the result is very close to achieving delta-neutrality. With normalized deltas with respect to the notional amount the mean delta exposure is of the magnitude 10^{-4} for the first case and 10^{-6} for the second. However, one thing to keep in mind is that everything is based on simulated values under some simplifying assumptions. This thesis should be a solid ground for future studies where more extreme scenarios are considered. With more extreme scenarios one could investigate the possibility to hedge with Gamma or another Greek such as Vega.

Sammanfattning

Med den ökande användningen av ränteswappar är det av stor vikt att förstå swaptioner, vilket är en option på en ränteswapp. En swaption kan användas både för spekulativa syften och för att hedgea mot risker i ränteförändringar. Det viktigaste att förstå är hur man prissätter en swaption. Eftersom swaptioner baseras på underliggande tillgångar så kommer vi börja med det mest grundläggande, obligationer, och arbeta oss fram till modellen vi kommer använda, Black76. Black76-modellen är en variant av den nobelprisvinnande modellen Black-Scholes-Merton. Med denna modell kan vi börja undersöka det huvudsakliga syftet med avhandlingen, en hedgningsstrategi för swaptioner från perspektivet av ett clearingshus.

Clearinghus är en central del av den moderna finansmarknaden. De agerar som en mellanhand för att hantera affärer mellan clearingmedlemmar och har en översikt över marknaden. Ifall en clearingmedlem går i konkurs, kommer clearinghuset att ta över portföljen med tillgångar. Clearinghuset kommer att hålla en auktion för att sälja av portföljen. De strävar efter att hålla auktionen så snabbt som möjligt och det sker generellt efter 5 till 15 dagar. Medan de har portföljen så är de exponerade mot riskerna i tillgångarna och därför är det av största vikt att kunna hedga tillgångarna. I den här avhandlingen har en strategi och en algoritm tagits fram för att använda delta-hedging för att uppnå delta-neutralitet under normala marknadsrörelser.

Vi kommer att undersöka två olika fall. Det första fallet är när portföljen tas över när det är lång tid kvar till swaptionens förfallodatum. Då kommer vi att använda forward swaps för att hedgea och för att minska antalet swappar kommer vi att använda kompression. Det andra fallet är när förfallodatumet uppnås under tiden som clearinghuset håller i portföljen. Dagarna innan förfallodatumet kommer vi hedgea med forward swaps med kompression. När datumet är nått så kommer vi istället att använda ränteswappar.

I båda fallen är resultaten nära att uppnå delta-neutralitet. Med normaliserade deltan med avseende på det nominella beloppet är medelvärdet av delta-exponeringen av magnituden 10^{-4} för det första fallet och 10^{-6} för det andra. Men, det är värt att komma ihåg att allting är baserat på simulerade värden under förenklade antaganden. Denna avhandling bör utgöra en bra grund för vidare studier där man kan undersöka mer extrema marknadsrörelser. Med extremare rörelser skulle man kunna undersöka hedgningsstrategier med andra Greker som till exempel Gamma och Vega.

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Abbreviations

CCP Clearinghouse / Central Counterparty

VaR Value at Risk

SPAN Standard Portfolio Value of Risk

ZCB Zero-coupon bond

IRS Interest rate swap

TIBOR Tokyo Interbank Offered Rate

STIBOR Stockholm Interbank Offered Rate

ITM In the money

ATM At the money

OTM Out of the money

BS Black-Scholes

MC Monte Carlo

GBM Geometric Brownian Motion

bp Basis point

1 Introduction

1.1 Background

The financial market is a huge and complex system to understand. In the modern world trades are mainly done with computers, compared to ancient times where you traded hand to hand. With the accessibility we have developed there are countless of trades being made each day without the traders ever meeting. Most people use online platforms to buy and sell assets such as stocks and relies on these platforms to work. These platforms are built upon complex computer systems that moves the money between the actors and can be seen as a digital hand to hand trade. This digital trading raises several opportunities but also questions and concerns we didn't have in ancient times. How is this managed and how do we know that both actors in a trade have the financial means to complete the deal?

An example when this was handled without care is the latest financial crisis in 2008. Bad loans were given to people who couldn't meet the requirements and eventually caused failures of large financial institutions until the market collapsed [1]. In the aftermath of the crisis came stricter restrictions of controlling the counterparty risk and risk management in general [2]. How does these platforms that are used for trading work? And how are they managing the risks? Nasdaq, which is the company that this thesis is written at, is one actor making digital trading possible.

Nasdaq is a global actor within the financial sector. It provides market places and develops the technology needed for these markets [3]. Nasdaq consist of several branches and the office in Umeå is part of the technology branch. More specifically, the team that this thesis is written with is responsible for the risk calculations in the technology that's being developed at Nasdaq. These risk calculations are limited to the instruments (derivates) that has been implemented in their system. One of the instruments that currently isn't supported is swaption. Swaptions are an option on interest rate swaps, which is an instrument becoming more and more common. With the increasing popularity of interest rate swaps, swaptions should also become more popular since they can be used both for speculation and hedging of interest rates.

Currently there are few companies that offer clearing possibilities of swaptions and those who does have monopoly on the instrument. However, new laws in Japan are making the clearing of swaps more restricted [4]. Swaps that are based on Japanese rates must be cleared through a local clearinghouse licensed by the Japanese government or through a foreign licensed clearinghouse. This means that participants on the swap market in Japan must trade their swaps in Japan, more or less. And the more trades there are in a market the more money there is to earn. Thus, one could speculate that other countries will follow Japan and limit swap trades to their own country. And if swaps increase then swaptions are likely to increase as well.

Since Nasdaq currently don't have swaptions implemented in their risk system, the knowledge of the instrument is insufficient. Therefore, the main scope of this thesis is to achieve a fundamental understanding of swaptions, from a clearinghouse perspective.

The most important aspect to understand is the valuation and pricing formula. With that knowledge the opportunity to investigate how a portfolio of swaptions will react to different market scenarios and how they can be hedged. Both aspects are of great importance to a clearinghouse. Why they are important will be easier to understand after the explanation of how a clearinghouse work.

1.2 Problem description

In order to create and offer software that support clearing of swaptions, the knowledge of the instrument must be increased, and the pricing model must be found. With the pricing model found, the parameters of the model can be investigated, and relevant scenarios can be created. The scenarios will be simulated with Monte Carlo and a hedging strategy will be constructed and tested. Since a clearinghouse doesn't know when they will receive a portfolio and only hold it for a short and limited time, the hedging strategy will be tested during two different periods of time. The different periods will lead to different approaches in the hedging strategy since the instrument changes during the life of it.

1.2.1 Goal and limitations

The goal of this thesis is to find a pricing model and create a hedging strategy that reduces the risk exposure as much as possible. One of the limitations is the lack of real data. Instead, the scenarios will be simulated with reasonable values and under certain simplifying assumptions. Another limitation is the knowledge of the subject. Swaptions are an exotic instrument and not commonly used. Therefore, there are limited amount of previous work done in the subject and inspirational articles are difficult to find.

2 Theory

This section will start with an introduction of clearinghouses and what defines them. Because the thesis is written from a clearinghouse perspective it's important to understand the parts of the clearinghouse and why the price model and hedging is important. When testing a model, the scenarios are a key factor. Therefore, the concept of stress tests and how to construct one properly will be covered. Before looking at swaptions, it's important to understand what mathematical background they are based on. Therefore, the start will begin with the basics, bonds, and move over to interest rate swaps before arriving at swaptions. There will also be an introduction to what options are and how to price them followed by compression. The basic idea of hedging and defining the Greeks will be covered before moving to the method.

2.1 Clearinghouse

A clearinghouse, also known as central counterparty (CCP), can be thought of as the middleman in financial transactions and is central to the modern world trading structure [2]. However, not everyone is allowed to trade directly through a CCP. This privilege is contained to the clearing members. To become a clearing member, you must be a large institution/company with a strong financial background. Usually, the members must pay an entrance fee and have good credit rating [2]. Smaller actors can trade with a clearing member and therefore indirectly use the clearinghouse [5]. To put the size of the members in perspective, a private person would typically turn to a smaller actor, which turns to a clearing member to use the CCP. The main purpose of a CCP is to reduce the counterparty risk [2]. Counterparty risk can be seen as the possibility that the counterpart in a trade don't have the financial means to meet the requirements of the contract. How CCP's reduces the risk will be covered more after the benefits of a CCP are more thoroughly explained.

The benefit of a clearinghouse is not only to reduce counterparty risk but also to simplify the clearing of trades. Due to the size of the financial market and the number of trades made each day the transactions between members becomes complex. Instead of having all the members settling every trade they can use the CCP to do that for them [5]. This results in dealing with one party instead of two, three or 500 different parties each day. The result of the reduction of transactions is illustrated in Fig(1). With all the transactions going through the CCP, the information about occurring trades is easier to overlook. This makes it easier to see if there are indications that some members are having financial problems, or we are heading towards a crisis.

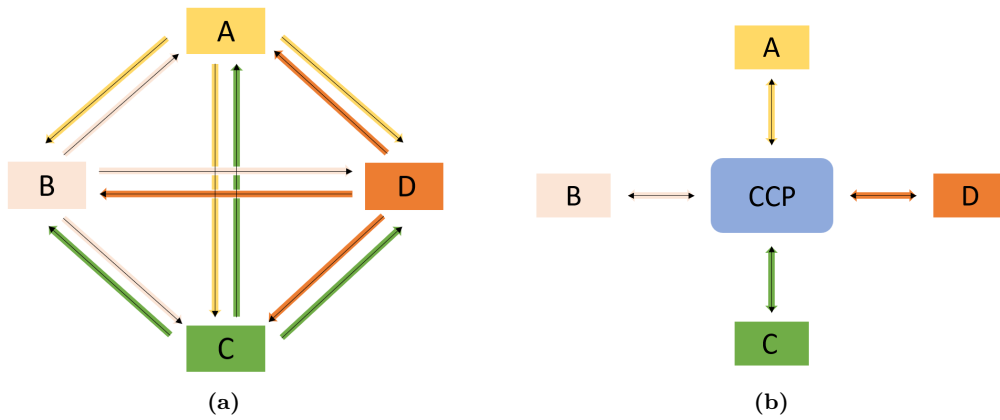


Figure 1: Illustration of four clearing members and how a CCP simplifies the transactions where (a) is without a CCP present and (b) with a CCP. By posing as the middleman and making the transactions go through the CCP the number of transactions can be reduced.

Something worth noticing is that the arrows in Fig(1b) goes in both directions. This indicates that there are some amounts of money going from A to CCP and some amount going from CCP to A. This means that in our example with four members, the number of transactions is reduced from twelve to eight. It's a decrease, but it can be reduced even more.

CCPs use something called multilateral netting [2] and this is illustrated in Fig(2). What this means is that the CCP calculates the net difference in the payments between the members. Instead of sending transaction back and forth they only transact the netted amount. With the presence of a CCP and usage of multilateral netting, the number of transactions, in this example, are reduced from twelve to four.

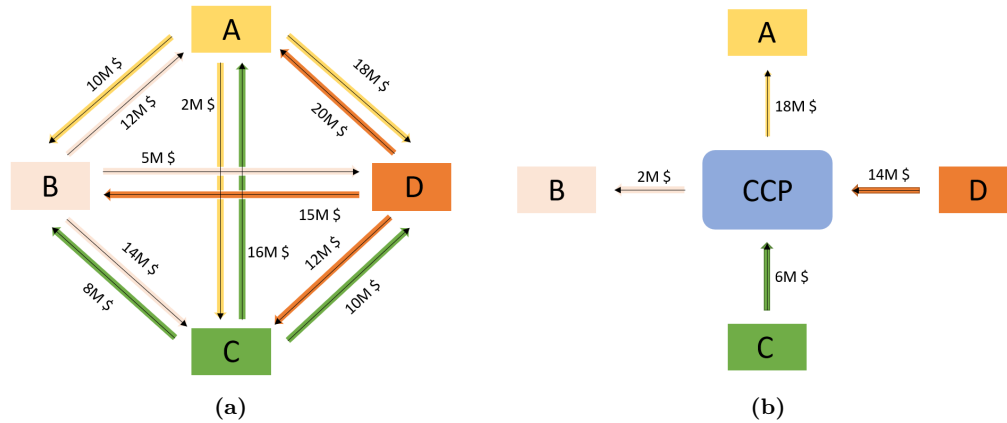


Figure 2: Illustration of how a CCP simplifies the transactions by posing as the middleman using multilateral netting. (a) is without multilateral netting and a CCP and (b) is with CCP and multilateral netting. By using multilateral netting the total amount a member should pay/receive is calculated and the transaction is reduced to one per member instead of three. In some cases, a member can become neutral after the netting and will have zero transactions.

2.1.1 Counterparty risk reduction

Now that some of the benefits of a CCP is explained, lets return to how they reduce counterparty risk. Since a CCP pose as a middleman, they act as both buyer and seller of contracts between the members. This means that if one member can't meet the requirements and declare bankruptcy, also called default, the CCP still have the obligation to complete the trade with the other member [2]. If a member defaults, it indicates that they are facing major losses on the contracts they have entered. The CCP will get ownership of the risk-filled portfolio and therefore have the same bad situation as the defaulted member. This position is not desirable and CCPs are trying to prevent this from happening, but how are they doing it? They use something called margin [2].

2.1.2 Margin

A margin is a form of collateral, or security, in form of financial assets such as cash, stocks, bonds, etc., but the preferred asset is cash. Margin can be divided into two parts, initial margin and variation margin. At the initialization of a contract, the CCP will demand both members, A & B, to post initial margin that reflects a percentage of what they stand to lose. Each day the CCP will perform risk calculations such as stress tests, which we will cover later, on the asset in the contract. Depending on what the calculation predicts, the CCP will demand more margin from the member on the losing side, B, and this is the variation margin. On the next day the market might have shifted, and the calculations shows that B seems to make a profit. Then B could withdraw the deposited

variation margin and A will need to post margin instead. Worth mentioning is that if a member post margin in forms of other assets than cash, there will be a haircut of that asset [2]. Meaning that since there is an uncertainty of a stock's value in the future the CCP will not count a stock that today is worth 1\$ as 1\$. If the haircut of the stock is 0.8 the member must post stocks for the value of 1.25\$.

2.1.3 Default waterfall

If a member is unable to post margin the CCP will put them in default and as mentioned earlier, receive ownership of the members portfolio. The first step to resolve the matter is to use the margin. Regardless of if the margin is enough, the CCP will hold an auction of the portfolio. This is because the CCP don't want to be a participant, only a middleman. The participants in the auction are the other clearing members. The auction will not be held instantly, but the CCP strive to hold it as soon as possible so they don't are exposed to any risks. Hopefully, the members are willing to buy the positions and the matter is resolved. However, if the margin and the money raised in the auction don't cover the losses there are several steps the CCP will take to clear the debt and complete the contracts the member had entered. These steps are part of the Default Waterfall [2] which contains "layers" of money and are described below. The order and the flow of the waterfall are illustrated in Fig(3).

- **Collateral**

The first step in the waterfall is called collateral and contains the money from the margin which was described above.

- **Skin in game**

This step is called skin in game because it refers to the CCP contributing with their own money to cover the losses. This is expected by the other clearing members to know that the CCP want the same result as the members. Mainly that everything should be run smoothly. It is the CCP's responsibility to perform high-quality risk estimations so that the margin is enough to cover the losses in case of a defaulting member. If the calculations are off or there is an unforeseen scenario, they will have to put up their own money to cover the mistake.

- **Default fund**

If the money in the above steps is not enough the largest layer of money, the default fund, is reached. The default fund is a pool of money where all the members in the clearinghouse have contributed. Usually, this contribution is the entrance fee they paid when becoming a member. As imaginable, the members won't be happy if this fund is depleted since it must be refilled by them and the CCP. This layer has only been insufficient a few times, and if this occur there are high chances that there will be severe consequences.

- **Right of assessment**

At this level the CCP hold the right to call the non-defaulting clearing members

to contribute with more money than they already did in the default fund. This is known as the rights of assessment. As mentioned, if this level is ever reached, we are in deep trouble. If the money still is insufficient, the CCP will put up the last of their equity before the last step of the waterfall is reached.

- **Last chance**

A bankruptcy of a clearinghouse would be devastating for a country's economy, therefore the last chance the CCP has is to ask the government to help them out. In most cases there is a smaller loss for the government to bail the CCP out rather than to face the consequences of a bankruptcy.

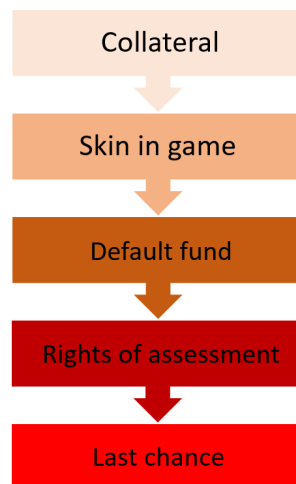


Figure 3: The steps of the default waterfall, i.e., the "layers" of money. The layers are put in the order from which the CCP use money to clear the debt of a defaulting member.

2.1.4 Stress test

In order to determine how much margin a member is required to post, CCPs uses traditional risk measurements such as VaR and SPAN, read more about them here [8]. A stress test is used to test how much of the waterfall would be consumed in case of one or several defaulting members. Stress tests are used to calculate and help manage the risks on the portfolios of the clearing members [2]. Stress tests are simulations with extreme but plausible scenarios that will examine how a portfolio reacts to hurtful events. These scenarios might be from historical data, such as the crash in 2008, or scenarios that are made up. In order to know how to construct a stress tests, the pricing models must be investigated. With the knowledge of the pricing model the parameters can be tested and the hurtful scenarios can be distinguished. For guidance on how to construct the scenarios for a stress test, see [6].

2.2 Bonds

A bond is similar to a loan, where the holder of the bond lend money to the bond issuer. As an example, an actor can issue multiple bonds to receive money that a single bank wouldn't give them. There are different types of bonds and the most basic one is called zero-coupon bond (ZCB).

2.2.1 Zero-coupon bond

In a ZCB the holder will pay the issuer at the beginning, t_0 , and receive a larger amount at an agreed time in the future. This agreed time in the future is called the maturity, T , of the bond. The amount paid at maturity is called the face value, N , and is larger due to the growth of money over time. The money will grow with the interest rate, r , and the price, $P(t_0, T)$, of the bond at time t_0 can be expressed as the present value of N

$$P(t_0, T) = \frac{N}{(1+r)^{(T-t_0)}},$$

where $1/(1+r)^{(T-t_0)}$ is the discrete discount factor. For simplicity the notation $B(t_0, T) = 1/(1+r)^{(T-t_0)}$ is used, which gives the final price for a ZCB as [7],

$$P(t_0, T) = B(t_0, T)N. \quad (1)$$

As mentioned, the ZCB is the simplest of bonds and now bonds containing coupon payments will be covered.

2.2.2 Coupon paying bond

A bond containing coupons is similar to a ZCB, where the holder will pay the issuer at the beginning and receive the face value at maturity. But the holder will also receive some defined percentage of the face value at certain intervals. These percentages of N are called coupons, denoted as c [7]. How often these coupons should be paid is defined in the contract and will vary from bond to bond. Important to notice is that the price of a bond with coupons are, just as for a ZCB, the present value of N but also the coupons. This means that the discounted value of the coupon at each payment interval must be added to the present value. The length of the intervals are expressed as $1/m$, meaning that for semi-annually (every 6 months) we have $m = 2$. Before going forward the discrete discount factor should be clarified again. For the ZCB, it looked like this

$$\frac{N}{(1+r)^{(T-t_0)}},$$

which is the case for an annual rate. Now that the interval, $1/m$, is introduced it must be added to the discount factor as well, giving us

$$B(t_0, T) = \frac{1}{(1 + \frac{r}{m})^{m(T-t_0)}}. \quad (2)$$

With that clarified, the present value of the bond with $1/m$ interval coupon payments can be expressed as

$$PV(t_0, T) = B(t_0, T)N + B(t_0, t_1)c_1 + B(t_0, t_2)c_2 + \dots + B(t_0, T)c_T,$$

which can be simplified into [7]

$$PV(t_0, T) = B(t_0, T)N + \sum_{i=1}^n B(t_0, t_i)c_i. \quad (3)$$

In the summation, n , represent the periods $n = T \cdot m$, meaning that $t_n = T$. This is because if the intervals are non-annual there will be more or less than one cashflow per year and that must be accounted for. As an example, with semi-annual there are twice as many cashflows compared to annual. As mentioned earlier, the coupons are a percentage of N , and this percentage can either be a fixed or a floating rate. From now on the fixed rate, or stationary rate, will be denoted as r_s and the floating rate as r_f . By rewriting Eq(3) with the definition of a coupon for a fixed rate, the present value, $PV_s(t_0, T)$, can be expressed as

$$PV_s(t_0, T) = B(t_0, T)N + \frac{r_s N}{m} \sum_{i=1}^n B(t_0, t_i). \quad (4)$$

The floating rate requires a bit more thought since the rate change for each period, but it's easier to calculate after some consideration. By starting simple, the first coupon payment with a floating rate can be described as

$$c_1 = \frac{r_f(1)N}{m},$$

where $r_f(1)$ can be expressed as the rate between time t_0 and t_1 as $r(t_0, t_1)$. This will be the case for all intervals $r_f(i) = r(t_{i-1}, t_i)$. For simplicity $m = 1$ is used. At maturity, T , the last payment will be the coupon payment plus the face value. Since it's the last time step the present value, at t_{T-1} , of the payment will be $B(t_{T-1}, T)(N + r_f(T)N)$. The floating rate at T can be expressed as $r_f(T) = r(t_{T-1}, T)$. Looking at Eq(2), $(T - t_{T-1}) = 1$ so the discount factor becomes

$$B(t_{T-1}, T) = \frac{1}{1 + r_f(T)} = \frac{1}{1 + r(t_{T-1}, T)}.$$

The present value at the last time step can then be expressed as [7]

$$\begin{aligned} PV_f(t_{T-1}, T) &= \frac{1}{1 + r(t_{T-1}, T)}(N + r(t_{T-1}, T)N) \\ &= N \left[\frac{1 + r(t_{T-1}, T)}{1 + r(t_{T-1}, T)} \right] = N. \end{aligned} \quad (5)$$

This will in fact hold for all time steps even if the step is bigger than 1 period because the rates can be rewritten as

$$(1 + r(t_0, t_1))(1 + r(t_1, t_2)) \dots (1 + r(t_{T-1}, T)) = (1 + r(t_0, T))^T,$$

which simplified becomes

$$\prod_{i=1}^n (1 + r(t_{i-1}, t_i)) = (1 + r(t_0, t_T))^T. \quad (6)$$

If m is 2 instead of 1, the interval (t_0, t_1) would represent half a year and the exponent in the discounting factor would still become 1 since $m \cdot 0.5 = 1$. This will hold for any interval such as quarterly, monthly and weekly. As previously mentioned, the floating rate r requires more thought but is easier since the present value will always be Equal to the face value.

2.3 Interest rate swap

Before moving into interest rate swap (IRS), let's clarify what a swap is. A swap is an exchange of an asset of any kind. This means that two actors can swap assets such as stocks, rates or perhaps apples for pears. In this thesis the focus will be on swaps of interest rates because it's very central for the progression towards understanding swap-tions. The most basic IRS is called plain vanilla interest rate swap where a fixed rate is swapped for a floating rate. Like a coupon paying bond, there are exchanges of cash flows on certain intervals. A cash flow is an exchange of money between the two traders. In an IRS there are two parts, the fixed leg and the floating leg. The fixed leg is the trader paying the fixed rate and intuitively the floating leg is the payer of the floating rate. From now fixed leg payer will be denoted as the payer and the floating leg as the receiver. When entering an IRS there are some formalities that must be set [7].

- **Notional amount**, N , - What is the amount we are calculating the cash flows on.
- **Maturity**, T , - How long is the contract going to be.
- **Payment intervals**, m , - How often are cash flows going to be exchanged.
- **Fixed rate**, r_s , - What is the fixed rate.
- **Floating rate**, r_f , - What is the floating rate based on.

The first three points are not worth mentioning since it's up to the traders to determine. The rates are however more interesting to talk about. The fixed rate is usually based on the yield curve. The yield curve is a curve that displays interest rates that have different maturities. It is used as a benchmark to match rates that for example banks are giving out. The longer the maturity is, the higher the rate is due to the risk associated with time. The floating rate is usually based on rates known as X Interbank Offered Rate. The X represents from what country/region the rates come from. For example, TIBOR stands for Tokyo Interbank Offered Rate and STIBOR is Stockholm Interbank Offered Rate. These "XIBORS" are calculated on collected reports from banks in a region. There are determined rates for different time periods such as 1month, 3months, 1year, 5year,

etc.

The TIBOR, which is used in this thesis, is updated each day so there will be new rates for 1M, 3M, 5Y, each day. So, in an IRS the payer rate is based on the yield curve and the receiver is based on TIBOR. Since the TIBOR is updated each day, at each new payment interval one must look what the new TIBOR is and use that as the receiver rate. Therefore, the receiver cash flow, at any interval (i), can be expressed as

$$cf_f(i) = \frac{r_f(i)N}{m}. \quad (7)$$

The payer rate will always stay the same so the payer cash flow can be expressed as

$$cf_s = \frac{r_s N}{m}. \quad (8)$$

Notice that these payments are very similar to the coupon payments for fixed and floating bonds. In fact, they are identical, you can create an IRS by combining one fixed and one floating bond with same notional, maturity and payment intervals. The present value, $PV_s(t_0, T)$ for the IRS from the payer perspective can be expressed as [7]

$$PV_s(t_0, T) = \frac{N}{m} \sum_{i=1}^n B(t_0, t_i)(r_f(i) - r_s). \quad (9)$$

When would an IRS be beneficial? Let's make an example.

A hotel is planning an expansion and borrows money from a bank with a floating rate over ten years based on TIBOR. After two years the owner of the hotel believes that TIBOR will increase significantly. Knowing that the hotel can't afford to pay more than 5%, the owner scouts the market for an IRS to swap the TIBOR for a fixed rate. Regardless of if the TIBOR would surpass 5% or not, the owner would be happy knowing that no matter what happens the expansion can continue according to plans. To further clarify an IRS, we consider the example above with values. Let's go back and define the properties of the IRS, see Table(1).

Table 1: The made-up properties for the IRS example.

N	10M\$
T	8 years
m	1 year
r_s	5%
r_f	TIBOR

The TIBOR for eight years is also needed, see Table(2)

Table 2: 1 year TIBOR for 8 years (made up numbers).

Year	1	2	3	4	5	6	7	8
Rate	3%	4%	5%	6%	7%	7%	4%	3%

With the values in Table(1&2) the fixed cashflows can be calculated as

$$cf_s = 5\%10M\$ = 0.5M\$$$

and the floating cashflows as

$$cf_f(i) = [0.3, 0.4, 0.5, 0.6, 0.7, 0.7, 0.4, 0.3]M\$.$$

The valuation of the IRS for the hotel owner can be expressed as

$$V_f = (cf_s - cf_f(i)) = [0.2, 0.1, 0, -0.1, -0.2, -0.2, 0.1, 0.2]M\$ = 0.1M\$.$$

In this scenario, the prediction of the hotel owner was correct, and a profit was made. An easier way to visualize the cash flows are with a schematic figure. In Fig(4a) the cashflows before netting are shown and in Fig(4b) the netted cashflows are shown.

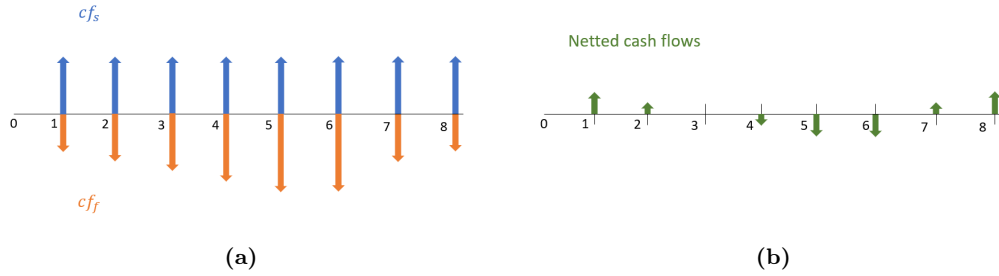


Figure 4: The cashflows where (a) shows the fixed and floating cashflows and (b) the netted amount for each interval.

As mentioned earlier, an IRS can be replicated by combining two fixed and floating coupon paying bonds. We can calculate the present value of a payer swap as the difference between the present values of the bonds, PV_s and PV_f . Using Eq(4) and the fact that $PV_f = N$ we get

$$PV_p(t_0, T) = N - \left(B(t_0, T)N + \frac{r_s N}{m} \sum_{i=1}^n B(t_0, t_i) \right). \quad (10)$$

Worth mentioning before moving on is that an IRS is not limited to swapping a fixed and float rate. A trader in Japan might be concerned that TIBOR will increase but believes that EURIBOR will remain low. Then a swap between two floating rates can be performed.

2.3.1 Forward swap

A forward swap is a contract to enter into an interest rate swap in a future point in time. Forward swaps will also be used and are very similar to a interest rate swap. The

difference is as mentioned, that the interest rate swap will start in a future point in time instead of directly. The present value formula for a payer forward swap is calculated as

$$V_{p.fwdswap} = \frac{N}{m} \sum_{i=0}^n B(t_0, T_i)(F_{r_T} - K). \quad (11)$$

2.4 Options

An option is a contract between two traders of one or multiple underlying assets [5]. In this thesis only two types of options are considered, call and put, and there are two positions, long/holder and short. A call option refers to buy the asset and a put to sell. What differs an option from a regular trade is that the holder of a call option, or the trader with a long position holds the right but not obligation to buy the asset. The other position in the call option is called short and they have agreed to sell the asset towards a premium, paid by the holder. In a put option the long position is the seller and the short is the buyer. If a trader with long position in a call/put option chooses to buy/sell it's called to exercise the option. To have the advantage of choosing whether to exercise or not, the trader with the long position will pay a premium to the short position at the initiation of the contract. How the premium is determined will be covered later. There are different kinds of call and put options which specifies when they can be exercised. The most common options are called European and American options [5]. In a European option you can only exercise at maturity but in an American you can exercise at any time during the contract. There are multiple other types of options, but the focus will be on European because it's known to be the easiest to understand and calculate.

For a European call option, with a stock as the underlying asset, the value is determined as the difference of the market price and the agreed price in the contract. If the price in the contract is lower than the price on the market the holder would exercise the option. If the price in the contract is higher than market the holder would let the option expire and instead buy directly from the market. The idea is the same for a put option but the other way around. Therefore, we can express the value of a long position in European call/put option as either zero or the positive difference between market and contract price as shown in eq(12) [5],

$$\begin{aligned} V_c &= \max[0, S - K], \\ V_p &= \max[0, K - S]. \end{aligned} \quad (12)$$

Where S is the market price of the stock and K is called the strike price, which is the agreed price of the stock in the contract. As seen in eq(12) the value of the option will always be non-negative. What is not considered in this valuation is the premium the holder has paid. So, the value of the option, if not exercised, is zero but the loss is contained to the premium. From the holder's perspective this is very safe since the loss is contained but the profit is theoretically unlimited. The benefit is shown in the following example.

Let's say at initiation, a stock is trading at 10\$ and the strike is set to 12\$ with maturity one year. The price (premium) of the option is 2\$ and at maturity the market price of the stock is 20\$. The value of the call option becomes $V_c = \max[0, S - K] = 8\$$ and our profit, P , is $P = V_c - \text{premium} = 6\$$. There is no telling how much the stock could increase in a year, which is why the profit could theoretically be unlimited. If the market price at maturity was 8\$ the value would be $V_c = 0$ and the profit $P = 0 - \text{premium} = -2\$$. In this scenario the holder made a loss on the investment. But regardless of how low the market value becomes, the loss is always limited to the premium 2\$.

The terms ITM, OTM and ATM is common terminology when talking about options and refers the status of market and strike price. The terms will be described from the perspective of a call option, for a put its mirrored [5].

- **ITM** - In the money. When the market spot price has surpassed the strike price, indicating there will be a profit.
- **OTM** - Out of the money. When the market spot price has not yet reached the strike price, indicating the option will not be exercised.
- **ATM** - At the money. When the market spot price is equal to the strike price.

In our example above, the first scenario would be described as ITM and the second scenario as OTM. Now we will continue to how we determine the price, or premium, of an option.

2.5 Option pricing

There are some ways of pricing an option and in this thesis the focus will be on the Nobel prize winning Black-Scholes model [7], (BS), and the Black76 model, which is a variant of the BS model [7].

2.5.1 Black-Scholes model

In 1960, options were traded without a pricing formula. The price was determined by intuition such that for a call option, the price increases with decreasing strike price, long maturity, high rate and volatility. But in 1973 the paper *The Pricing of Options and Corporate Liabilities* written by Fischer Black and Myron Scholes with acknowledgement of Robert Merton was published. After the paper was published it became the most used model when pricing options. There are however some assumptions being done in this model which are important to mention [7].

- **Arbitrage** - There are no risk-free arbitrage opportunities.
- **Transactions** - There are no transaction costs.

- **Risk free rate** - The risk-free rate is constant.
- **GBM** - The stock prices are random and follows a geometric Brownian motion.
- **Stocks** - The stock prices are continuous and pay no dividends.
- **Volatility** - The volatility is known and constant during the period.

With these assumptions, according to BS, the formula for determine the price of a European call option is eq(13) [7]

$$C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2). \quad (13)$$

S is the stock price today, K the strike price and e^{-rT} the continuously compounded discount factor. Previously when working with rates, the discrete discounting was used since there was distinct intervals. A stock can have updates multiple times a day and therefore the continuously compounded discount factor is used. The function Φ is the cumulative normal distribution functions. The variables d_1 and d_2 are calculated as [7]

$$\begin{aligned} d_1 &= \frac{\ln(\frac{S}{K}) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} \\ d_2 &= \frac{\ln(\frac{S}{K}) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}. \end{aligned} \quad (14)$$

Where σ is the volatility, T the number of trading days left to maturity and r the risk-free rate.

2.5.2 Black76 model

As mentioned earlier, Black76 is a variant of BS where instead of a stock the future price, F, of the underlying asset [7] is used. The Black76 model is used when working with contract on futures or instruments that are based on a forward curves. F must be estimated, which can be done in several ways. In this thesis Monte Carlo, which is covered in the method, will be used to predict the value. In eq(13) the discount factor was only used on the strike price since S is the price of the stock at current day. Therefore, the pricing formula for a call/put option on a future price is calculated as eq(15) [7]

$$\begin{aligned} C &= e^{-rT}[F\Phi(d_1) - K\Phi(d_2)] \\ P &= e^{-rT}[K\Phi(-d_2) - F\Phi(-d_1)]. \end{aligned} \quad (15)$$

With d_1 and d_2 as eq(16) [7]

$$\begin{aligned} d_1 &= \frac{\ln(\frac{F}{K}) + \frac{\sigma^2}{2}T}{\sigma\sqrt{T}} \\ d_2 &= \frac{\ln(\frac{F}{K}) - \frac{\sigma^2}{2}T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}. \end{aligned} \quad (16)$$

With this theory covered, let's start looking at swaptions.

2.6 Swaptions

A swaption is an option on a swap. As mentioned earlier the focus will be on European swaptions, which means at maturity the holder has the right but not obligation to exercise the option and enter the underlying swap. Recall that in an option there are two types, call and put. The payer of the fixed rate is synonym with a call option and is referred to as payer swaption [8]. The receiver of the fixed rate, and payer of the floating rate, is called receiver swaption and is thought of as the put. Our point of view will be the payer swaption.

When are swaptions desirable? Let's revisit the example of a hotel expansion, but this time using a swaption instead of an IRS. The hotel owner takes a loan from a bank with floating rate based on TIBOR. The owner knows that they can't afford to pay more than 5% in rate and therefore scouts the market again, but this time for a swaption. The owner takes a long position in a payer swaption with the same maturity and notional as the loan, with a strike rate of 5%. The owner has now secured the right to pay 5% and receive TIBOR. The difference here from the IRS is that at t_0 when the swaption is initiated nothing happens. In a swaption we have the initiation date, t_0 , maturity T , and the expiration of the underlying swap at T^* . Previously it was stated that the maturity is when the option expires, and this is still true. However, in a swaption the underlying swap begins when the swaption expires. This means that the maturity of the swaption might be in two years and then the rate swap activates for another eight years, if the option is exercised. After two years when the swaption is about to expire the owner will check the market for the value of eight-year fixed rates. If fixed rates are trading at 6% the owner is satisfied with paying 5% according to the swaption contract and chooses to exercise the swaption. The expansion plans can continue because the rate is locked for 5% for the remaining time of the loan. If the rates are instead trading at 4% the owner will not exercise the option, but instead enter an IRS paying 4% as fixed rate, see Figure [5] for explanatory figure. Either way the expansion can continue, and the owner is happy.

What differs from the case with an IRS is that the hotel owner has a chance to make money if the swaption is exercised. If market rates are trading at 6% and the owner's rate is locked at 5%, the owner could enter a receiver IRS with a different trader. This would mean that the owner receives TIBOR from the swaption and pays TIBOR in the IRS and these two are netted to zero. Now to where the money is made. The owner would pay 5% fixed in the swaption but receive 6% from the IRS making a profit of 1% each cashflow without any risk since the floating rates are netted against each other.

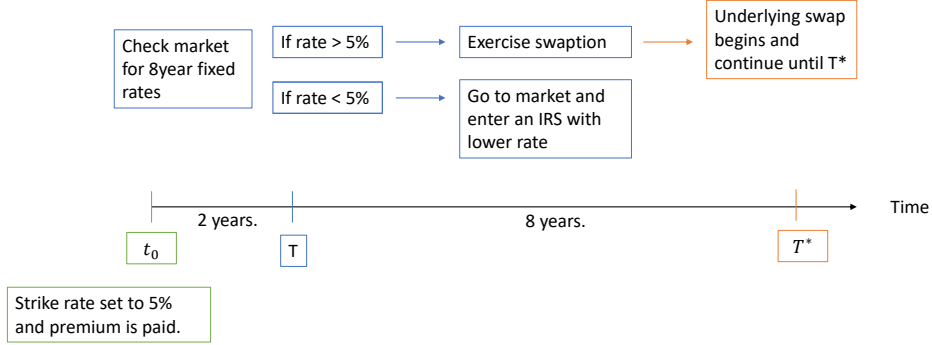


Figure 5: The timeline and explanatory choices for the life of a swaption. At the initiation of the contract, at t_0 , the conditions of the contract are set, and the premium is paid. After 2 years, at maturity T , the swaption will either be exercised or not. If exercised, the underlying interest swap will begin. The maturity of the interest rate swap is at T^*

2.6.1 Pricing swaptions.

When pricing a swaption the future price, F , or in this case the future rate, F_r , must be estimated. How this is done is covered in the method, for now it's enough to know that it's estimated with Monte Carlo. As mentioned in the hotel example, there are three points in time that needs to be considered. t_0 , the initiation date of the swaption contract. T , the maturity of the swaption and the start of the underlying swap contract. T^* , the maturity of the underlying swap contract. At t_0 , F_r must be estimated at T in order to price the swaption. The estimation of F_r begins by looking at the present value of the swap contract at time T . The derivation follows the combined steps from [7] and [8]. Changing the time in Eq (10) gives

$$PV_T = N - \left(B(T, T^*)N + \frac{r_s N}{m} \sum_{i=0}^n B(T, t_i) \right). \quad (17)$$

Just as before n is used to describe the number of periods. The difference is that instead of the interval t_0 to $T \cdot m$, now T to $T^* \cdot m$ is used. Meaning that $i = 0$ is the same as $t_0 = T$. This is because the cashflows will only occur if the swaption is exercised and there is an active IRS from time $T \rightarrow T^*$. Setting the present value to zero and solve for

r_s gives

$$r_s \frac{1}{m} \sum_{i=0}^n B(T, t_i) = 1 - B(T, T^*). \quad (18)$$

To estimate the discount factors, recall Eq(6), but using continuously compounded discounting instead. With the same principle receiving

$$B(t_0, t_1)B(t_1, t_2) = B(t_0, t_2) \iff B(t_1, t_2) = \frac{B(t_0, t_2)}{B(t_0, t_1)} \quad (19)$$

to estimate the discount factors in Eq(18). Using the fact that $\frac{B(t_0, T)}{B(t_0, T)} = 1$, Eq(18) is multiplied with 1 giving us

$$r_s \frac{1}{m} \sum_{i=0}^n B(T, t_i) \frac{B(t_0, T)}{B(t_0, T)} = \frac{B(t_0, T)}{B(t_0, T)} - B(T, T^*) \frac{B(t_0, T)}{B(t_0, T)}. \quad (20)$$

Using the logarithmic laws, the discount terms can be rewritten. Only one term will be shown because the others are rewritten with the same procedure.

$$\begin{aligned} B(T, T^*) \frac{B(t_0, T)}{B(t_0, T)} &= e^{-r(T^*-T)} \frac{e^{-r(T-t_0)}}{B(t_0, T)} \\ &\iff \\ \frac{e^{-r(T^*-T)-r(T-t_0)}}{B(t_0, T)} &= \frac{e^{-r(T^*-t_0)}}{B(t_0, T)} = \frac{B(t_0, T^*)}{B(t_0, T)} \end{aligned}$$

Using this Eq(20) can be rewritten into

$$r_s \frac{1}{m} \sum_{i=0}^n \frac{B(t_0, t_i)}{B(t_0, T)} = \frac{B(t_0, T)}{B(t_0, T)} - \frac{B(t_0, T^*)}{B(t_0, T)}. \quad (21)$$

Now that the discount factors are rewritten to start at time t_0 , the yield curve can be used where fixed time rates are available. With this knowledge the future rate F_r at time T can be estimated. By multiplying all terms in Eq(21) with $B(t_0, T)$ and replacing r_s with F_{r_T} , the future rate can be expressed as

$$F_{r_T} \frac{1}{m} \sum_{i=0}^n B(t_0, t_i) = B(t_0, T) - B(t_0, T^*). \quad (22)$$

The value of a payer swaption is determined as the discounted difference between F_{r_T} and K as

$$V_{p.swaption}(T, T^*) = \frac{N}{m} \max[0, F_{r_T} - K] \sum_{i=0}^n B(T, t_i). \quad (23)$$

Now recall eq(15) which is discounted with time left to maturity. Since the swaption will produce a series of cash flows, the discounts for each interval must be added as well.

By doing this and using logarithmic laws, we arrive with the pricing formula for a payer swaption (call) as

$$P_{p.swaption} = \frac{N}{m} \sum_{i=0}^n B(t_0, t_i) [F_{r_T} \Phi(d_1) - K \Phi(d_2)]. \quad (24)$$

d_1 and d_2 are very similar to Eq(16) with a change of notation from F to F_{r_T} , we get

$$\begin{aligned} d_1 &= \frac{\ln\left(\frac{F_{r_T}}{K}\right) + \frac{\sigma^2}{2}T}{\sigma\sqrt{T}} \\ d_2 &= \frac{\ln\left(\frac{F_{r_T}}{K}\right) - \frac{\sigma^2}{2}T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}. \end{aligned} \quad (25)$$

For the price of a receiver swaption (put), the places on F_{r_T} and K are switched and a minus sign is added to the values inside $\Phi()$ giving us

$$P_{r.swaption} = \frac{N}{m} \sum_{i=0}^n B(t_0, t_i) [K \Phi(-d_2) - F_{r_T} \Phi(-d_1)]. \quad (26)$$

2.7 Hedging

The term hedging is frequently used within the world of finance and can seem as an intimidating and complex area to understand. It is actually a very complex area, but the main idea of hedging is however quite simple. The idea is to reduce your risk or exposure to certain events. An example could be as simple as a person taking the bike to work during the winter. A way to reduce your risk for injuries is to buy studded tires to reduce the exposure to icy roads. Another example could be a manufacturer of apple cider. The manufacturer is exposed to increasing prices of apples and can reduce the risk by entering a contract with a fixed price on the apples.

In the world of finance, the risk and exposures depend on what type of instrument we consider. Some examples are options, futures, stocks and bonds. These financial instruments are correlated to several factors such as underlying asset, expiry/maturity, volatility and many more. The focus will be on hedging swaptions and swaps and the factors that will be considered are the changes of the swaps underlying parameters, i.e., fixed and floating rate. This is because a swaption is an option to enter an IRS and the exposure are to changes in the interest rates. To help us calculate the exposure, there are a ton of quantities representing the sensitivity of the price to a change of the underlying parameter. These are called "The Greeks" [8].

2.8 The Greeks

The Greeks measures the sensitivity of an option, or portfolio, with respect to a change in an underlying parameter. Each Greek will provide the exposure to a certain parameter.

The first and most common Greek is called delta, Δ , which describes the change of the option value with respect to changes of the underlying asset. To calculate the Greeks, the Black76 pricing model for a swaption is used. By taking the partial derivative with respect to the underlying parameter that is investigated, in this case the future rate, the Greek in question is found. Delta is calculated as [8]

$$\Delta = \frac{\partial V}{\partial F_{r_T}}, \quad (27)$$

where V is the value of the option and F_{r_T} is the future rate at time T . By taking the partial derivative of the value of a payer swaption, eq(24), with respect to F_{r_T} the expression for $\Delta_{p.swaption}$ is found as

$$\Delta_{p.swaption} = NA\Phi(d_1), \quad (28)$$

with $A = \frac{N}{m} \sum_{i=0}^n B(t_0, t_i)$. When investigating how to delta hedge swaptions, the delta of a swap is also needed. Since the Greeks are for options, the same derivative can't be used for a swap. Instead, the finite difference method is used. There are three types, forward, backward and central. The forward difference will be used in this thesis. The forward finite difference is calculated as [8]

$$f'(x) = \frac{f(x+h) - f(x)}{h}. \quad (29)$$

This equation can be rewritten with our term to make more sense. $f(x)$ is rewritten as the present value of a payer swap, eq(9). $f(x+h)$ is the present value of a payer swap with the floating rate curve shifted with an amount h . $f'(x)$ will then represent the delta of a swap, $\Delta_{p.swap}$ and with these changes (29) can be rewritten as eq(30)

$$\Delta_{p.swap} = \frac{PV_{p.swap_{shifted}} - PV_{p.swap}}{h} \quad (30)$$

3 Method

In this section there will first be an explanation to how the rates are simulated and updated. This will be followed by the implementation of my hedging strategy. For all simulations, the made up values in Table(3) are used. These values were chosen because they seemed reasonable after looking at some fractions of real data.

Table 3: *Values for the simulations.*

Parameter	Meaning	Value
F_0	Initial rate	5%
K	Strike rate	[4.8, 5.0, 5.2]%
r	Discount rate	3%
σ	Volatility	20%
N	Notional amount	100000
M	Length of a year	360
m	Semi-annual payment interval	2 (=6 months)
i	Iterations to generate rates	20000
T	Swaption maturity	2 years
T^*	Underlying swap maturity, from t_0	5 years

3.1 Rates

Before we continue to delta hedging, let's look at the rates we will be using. For all simulations, the values in Table(3) are used. In this thesis the rates that we use to calculate the parameters of the instruments are simulated using Monte Carlo (MC). The idea of MC is to approximate the future by iterating lots of times and taking the average value [9]. In our case we simulate 20,000 rates which all start on the same value, but each rate is following a Geometric Brownian Motion (GBM). GBM was chosen because it was recommended by [7]. For those that want more information about GBM, see [11]. In Fig(6a) we can see how we first simulate 20,000 rates, all starting at 5% with stochastic evolution. In Fig(6b) we see the average rate of each day, calculated from the 20,000 rates. We simulate the floating rate by taking the daily fixed rate as starting point. Then we shift the fixed rate with a number between -3 to 1 bp. This is because generally the floating rate is lower than the fixed. As we can see in Fig(6c) the floating rate is below the fixed for most days. To calculate the values and deltas of the swaps we are using to hedge, we need to calculate the predicted values of TIBOR. The simulated TIBOR is calculated from the floating rate. If our interval is semi-annually, ($m=2$), TIBOR 6M is found by taking the rate with interval 180 days from maturity, T , up to T^* .

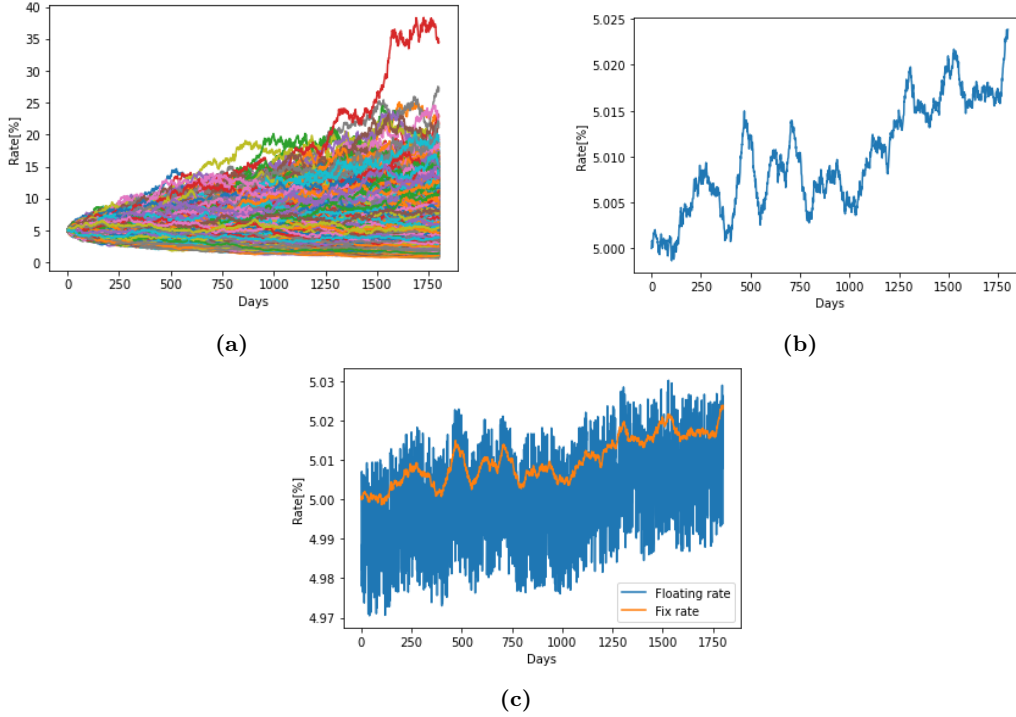


Figure 6: The process of finding our rates. In (a) 20,000 fixed rates are simulated with 5% as starting rate. In (b) the average fixed rate is calculated by taking the average of the 20,000 rates. In (c) the floating rate is simulated by for each day, the value of the fixed rate is added with a random number between $(-0.0003, 0.0001)$.

The result will be more accurate with higher number of iterations. Ideally you want a higher number of iterations than 20,000 but after a certain point the improvement will stagnate, but the computational cost will keep increasing. In Fig(7) we can see the difference in swaption price using the Black76 model and MC using different number of iterations. Since the MC is following the Black76 price closely enough for 20,000 iterations, we consider the slight improvement of increasing the iterations not worth the increase in computational cost. Therefore, we will continue using 20,000 iterations for the remaining simulations. We can also see how the price of the swaptions goes towards zero when we are approaching maturity. This is expected since it will be easier to predict if the swaption will be exercised or not the closer we get to maturity.

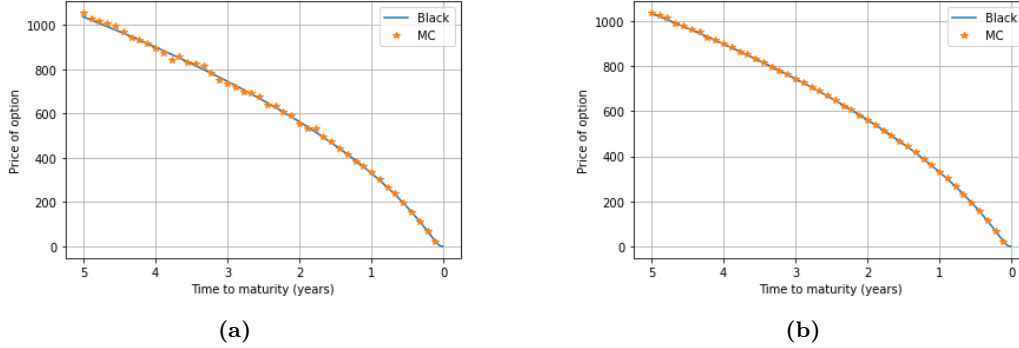


Figure 7: Comparing swaption price when using Black76 model and MC with varying number of iterations. (a) 20,000 iterations and (b) 200,000 iterations.

3.1.1 Rate updates

When performing our hedging strategy, the swaption maturity, T , is set to 2 years. The maturity of the underlying IRS, T^* , is set to 3 years starting at T , meaning that seen from t_0 we have a total length of 5 years. In case 1, when maturity is not reached before the auction is held, we will receive the portfolio 400 days from t_0 . This means that we will have 320 days until maturity is reached. The discount factors will be shifted accordingly and then our hedging strategy will begin. The following day, $t = 401$, we will simulate new fixed, floating and TIBOR rates with a starting value of the fixed rate at $t = 401$. These steps will be repeated until the auction is held. The strategy is not limited to these maturities and time when we get the portfolio, it's just to be consequent between scenarios. The steps are summarized below

- Day 1 ($t = 400$)
 - Use fixed, floating and TIBOR rates at day 320 to evaluate swaption and hedging swaps, also adjust discount vectors.
- Day 2($t = 401$) to day of the auction
 - Use fixed rate from day 401 as starting value and simulate new fixed, floating and TIBOR rates from day 401 to day 1800.
 - Now maturity is in 319 days. Use the rates of day 319 to evaluate swaption, hedging swaps, and adjust discount vectors.

3.2 Different cases

From a CCP's perspective, hedging of portfolios are very important. Recall from earlier that if a member defaults, the CCP will take over the portfolio. This portfolio will then

be sold at an auction that the CCP will host for the other members. This auction is typically held after 5 to 15 days, which means that the CCP is exposed to the risk of the portfolio. The aim is to hedge the portfolio during that period.

Our hedging strategy will be different depending on when we take over the defaulted members portfolio. Recall the timeline in Fig(5), if we get the portfolio long before T , the maturity of the swaption, we will be hedging an option. But what if we get the portfolio a few days before T ? Assume that before the auction is held, the swaption will reach its maturity. Depending on if the swaption is ITM or OTM we will have an IRS or nothing to hedge at all. If the swaption is ATM we assume that the holder will exercise the swaption. In this thesis we will consider the following cases, with h as the time the CCP will hold the portfolio.

- **Case 1:** $(t+h) < T$ - The maturity of the swaption is not reached during the time the CCP is holding the portfolio.
- **Case 2:** $t < T < (t+h)$ - The maturity of the swaption is reached during the time the CCP is holding the portfolio, giving us the following two scenarios.
 - ITM/ATM - The swaption is in/at the money, at T , and will be exercised.
 - OTM - The swaption is out of the money, at T , and will not be exercised.

3.2.1 Intuitive way of hedging swaptions and swaps

In this section we will cover the intuitive way hedging our cases before moving into the more detailed way of delta hedging. For the first case, when T is not reached, we will hedge the swaption with a forward swap. A forward swap is very similar to a swaption. At initiation a strike rate and the date when the swap should start is determined, as it is for a swaption. Like a swaption, we will estimate the fixed market rate at the start of the swap. If we compare the value of the forward swap, Eq(11) with the value of a swaption, Eq(23), we see that the only difference is that the value can become negative in the forward swap. Intuitively, the risk of a swaption would be neutralized by taking the opposite position in a forward swap with the same underlying parameters.

In the second case, when T is reached during the hedging interval, it becomes a bit trickier. Let's assume that the auction will be held after ten days, and we receive the portfolio with five days left to T . For the first five days, we would use forward swaps to hedge with the same intuitive motivation as in the first case. After five days when T is reached, we will instead of having a portfolio with swaptions either have a portfolio of active IRSs or nothing at all depending on ITM or OTM. If the swaption was ITM, and therefore exercised, we must hedge the active IRSs which is done by entering an IRS in the opposite leg. If the swaption was OTM, and not exercised, we must hedge our now active forward swaps that was initially meant to hedge the swaptions. When the forward swaps are active, they are IRSs so the strategy for this case will also be to entering an IRS in the opposite leg.

3.3 Delta hedge

Traditionally when talking about delta, you consider the value to be between 0 and 1 for call options and -1 and 0 for puts. If we look at Eq(28) we can see that the values will not be in that range, since a notional amount has no limits. These deltas will instead describe how much the option/swap will increase/decrease in value if the rate is shifted 1 bp. 1 bp is equal to 0.01% so if the rate is 5%, a 1 bp increase would result in a rate at 5.01%. In order to keep the portfolio delta neutral, we want to buy/sell certain amounts of forward swaps before T and IRSs after T so that the portfolio delta is as close to zero as possible. Usually, you want a hedge portfolio to be self-financing, meaning that you "borrow" the money you spend on your hedging instruments. For example, if you buy a swap to hedge a swaption, you get the money to buy the swap from investing in risk-free assets such as a ZCB. In our case as a CCP, we have money in-house from margin calls and the default waterfall that will finance the hedging. For more information about self-financing portfolios, see [11].

An important thing to clarify is as we recall from the section about options, the holder of the option only stands to lose the premium which is already paid. It is the trader in the short position that holds the risk, since the only payoff is the premium, but the loss is theoretically unlimited. The strategy we will perform is based on the holder's perspective in a payer swaption. However, the amount the short position stands to lose is just the negative value of the holder's profit.

3.3.1 Delta hedging with forward swaps

For the first case, and the first period in the second case, we will hedge the portfolio by going in opposite direction in forward swaps. What we want to achieve is

$$w\Delta_{p.swap} + \Delta_{p.swaption} = 0, \quad (31)$$

where w is the ratio of forward swaps we need in order to become delta neutral. w is calculated by rearranging the terms, and can be expressed as

$$w = -\frac{\Delta_{p.swaption}}{\Delta_{p.swap}}. \quad (32)$$

This ratio will rarely become an integer, and since you can't buy "half a swap" we instead adjust the notional of the swap, N_{adj} , so that $\Delta_{p.swap_{adj}}$ satisfies Eq(31). We calculate N_{adj} as

$$N_{adj} = wN. \quad (33)$$

When we have acquired N_{adj} we calculate $\Delta_{p.swap_{adj}}$ with Eq(30) which now should be $\Delta_{p.swap_{adj}} \approx \Delta_{p.swaption}$.

Our strategy will be to calculate the steps described above at the end of the day, make a buy/sell order overnight, and receive our hedging swap at opening the next

morning. At the start of the second day, the rates might have changed. Meaning that $\Delta_{p.swap}(t_1) \neq \Delta_{p.swap}(t_2)$, which implies that we won't be exactly delta neutral since $\Delta_{p.swap_{adj}}(t_1)$ was calculated to counter $\Delta_{p.swap}(t_1)$. At the end of the second day, we first calculate $\Delta_{p.swap_{adj}}(t_2) = \Delta_{p.swap}(t_2) - \Delta_{p.swap_{adj}}(t_1) \neq 0$. We calculate $\Delta_{p.swap}(t_2)$ with N and find the ratio w , but this time as

$$w = \frac{\Delta_{p.swap_{adj}}(t_2)}{\Delta_{p.swap}(t_2)} \quad (34)$$

and then we calculate $\Delta_{p.swap_{adj}}(t_2)$ which we place an overnight order on. These steps are repeated each day until the auction is held. The algorithm is summarized below where $i = 1, 2, \dots, n$ with n as the day of the auction.

- Day 1 ($i = 1$)
 - Simulate rates.
 - Calculate $\Delta_{p.swap}(t_1)$ and $\Delta_{p.swap}(t_1)$ with Eq(28) and Eq(30).
 - Calculate w with eq(32) and N_{adj} with Eq(33).
 - Calculate $\Delta_{p.swap_{adj}}(t_i)$ with Eq(30) and place overnight order.
- Day 2 to the day before auction ($i = 2, \dots, n - 1$)
 - Simulate rates.
 - Calculate $\Delta_{p.swap}(t_i)$ and $\Delta_{p.swap}(t_i)$ with Eq(28) and Eq(30).
 - Calculate $\Delta_{p.swap_{adj}}(t_i) = \Delta_{p.swap}(t_i) - \sum_{k=1}^{i-1} \Delta_{p.swap_{adj}}(t_k)$
 - Calculate w with Eq(34) and N_{adj} with Eq(33).
 - Calculate $\Delta_{p.swap_{adj}}(t_i)$ with Eq(30) and place overnight order.
- Day of the auction ($i = n$)
 - Simulate rates.
 - Calculate $\Delta_{p.swap}(t_n)$ with Eq(28)
 - Calculate $\Delta_{p.swap_{adj}}(t_n) = \Delta_{p.swap}(t_n) - \sum_{k=1}^{i-1} \Delta_{p.swap_{adj}}(t_k)$

3.3.2 Delta hedging with swaps

For the second period of the second case, we will hedge our portfolio with IRSs. If the swaption was ITM we will now have a portfolio with active IRSs, both from the swaptions and the forward swaps we bought to hedge. If the swaption was OTM we will only have active IRSs from the forward swaps. The hedging strategy will be very similar whatever case we find ourselves in.

- ITM

First, we calculate the delta of the IRS which the swaption was based on with Eq(30), giving us Δ_{swap} . Then we subtract the deltas from our hedging swaps, $\sum \Delta_{p.swap_{adj}}$, giving us $\Delta_{swap_{hedged}}$. To find the notional of the hedging swap, $\Delta_{swap_{adj}}$, we look at Eq(30). Since we want $\Delta_{swap_{hedged}} = \Delta_{swap_{adj}}$, we can use Eq(30) with $\Delta_{swap_{hedged}}$. By writing the present values in Eq(30) as in Eq(9) we can solve for N as

$$N = \frac{h\Delta_{swap_{hedged}}}{\left(\frac{1}{m} \sum_{i=0}^n (r_{s_i} - (r_{f_i} + h)) - \frac{1}{m} \sum_{i=0}^n (r_{s_i} - r_{f_i})\right)}. \quad (35)$$

With the notional calculated we place an overnight order on a swap. These steps are repeated until the auction is held.

- OTM

Since the swaption expired without being exercised, our only exposure is to the portfolio of forward swaps we bought to hedge the swaption. The delta we now want to neutralize is sum of the forward swaps purchased before maturity was reached. To find an IRS that neutralizes our exposure we use Eq(35) with $\Delta_{fwdswaps}$ and place an overnight order. The following day the rates might have changed and in that case, we find a new IRS with Eq(35) and this step is repeated until the auction is held.

3.4 Compression

If we continue the assumption that the CCP will hold the portfolio for ten days, with our strategy it will result buying/selling ten forward swaps/IRSs to hedge the swaption. A portfolio from a defaulting member might contain $1 \rightarrow 5000$ swaptions or even more. The conclusion we can draw from this is that the hedging portfolios will become massive. What if we could use something to reduce this amount? Recall the explanation of multilateral netting and its purpose from the theory section. In this case we can use compression, or advanced netting, to reduce the number of hedging instruments. A simple example is illustrated in Fig(8). A member, A, has a portfolio of two IRSs, both based on the same floating leg. The cashflow of the fixed leg from each swap is calculated and the netted amount is found. By reducing the notional amount and increasing the fixed rate we see that the same cashflow from the fixed leg is acquired. This can come very handy since we are using forward swaps to hedge our swaptions. The hedging swaps will always have the same floating leg but different notional and fixed rates. The limitation of this method is that the swaps must be based on the same floating leg. Meaning that we can't compress two swaps that are based on different XIBORS or have different payment intervals. A portfolio of eight swaptions could have two based on TIBOR 6M, four on STIBOR 3M and two on TIBOR 9M. There is however no limitation to how many swaps of the same type we can compress. This would mean that our hedging swaps could be reduced to a total of three swaps, one for each floating rate, instead of eight.

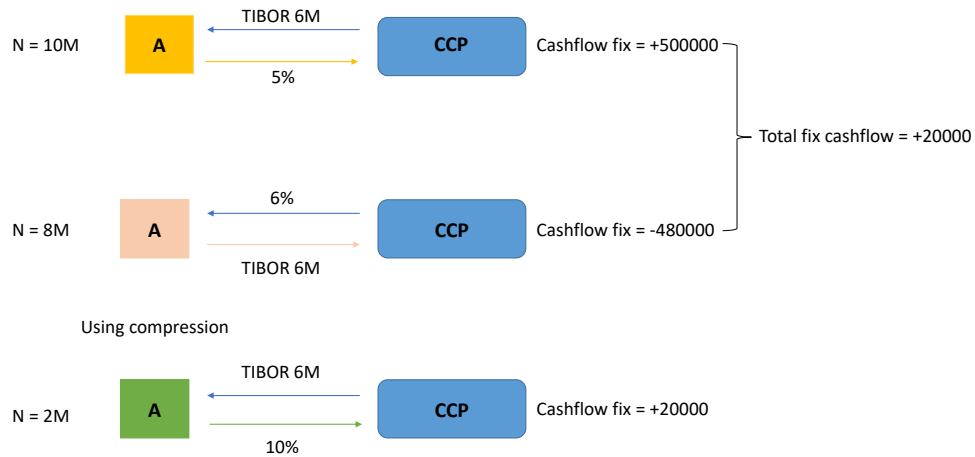


Figure 8: In order to reduce the number of swaps a member possesses, compression can be used. If the floating rate is the same for both swaps, the cashflows from the fixed legs can be calculated. If the fixed cashflows are in the same direction or in the opposite directions, as in this figure, the notional amount is added or subtracted. With the new notional the fixed rate is adjusted so that the total cashflow from the two swaps are equal to the new compressed swap. By doing this we have not changed any cashflows, only reduced the number of swaps a member possesses.

4 Result and Discussion

In this section the results of the hedging strategies for case 1 and case 2 will be presented. To be able to get a distribution, both cases were simulated 10000 times. The length of the hedging interval, i.e., how long the CCP hypothetically holds the portfolio, is ten days. As mentioned earlier, all simulations used the values in Table(3). The only variation is the strike rate, K , to see what happens if we begin the simulation by being OTM, ATM or ITM. It is important to remember that even if we begin OTM, ATM or ITM that does not guarantee that we will stay that way when receiving the portfolio. Theoretically we could end up OTM for all 10000 scenarios even if we began ITM. But for most cases, we should end up as we began. For case 1, the state we are in when we receive the portfolio should matter less than for case 2. This is because there is still a long time left to maturity and the outcome is still very uncertain.

4.1 Case 1

For case 1 we receive the portfolio after 400 days, with 320 days left to swaption maturity. For each simulation, the delta exposure of the last day of hedging was saved and in Fig(9) we see the distribution of the simulations. In the distributions we can see that we are very close to being delta neutral, with the means, μ , in the interval $(-3.14e-4, 8.39 e-4)$. The results in Fig(9) are normalized with respect to the notional amount. We can interpret the results as the following. Without the normalization we would see the x-axis with values such as -1000, 1000, 2000, etc. If the delta is 1000, a shift of 1 bp would mean an increase of 1000 crowns. Similarly if delta is -1000 but a decrease. Why we are seeing a spread is because of the hedging updates. Since the swap we purchase to hedge day one, will hedge for day two, we will have a delay and the last hedging swap will not be purchased the day the auction is held. By visual inspection of the distributions and comparing the means and standard deviations, σ , we see that starting ITM, ATM or OTM does not have a significant effect on the hedging performance. As I briefly mentioned earlier, this was expected and most likely it is because there is still a long time left to maturity. Therefore, there are not so much to discuss about the distributions since they look very similar no matter how we start.

However, what can be discussed are the scenarios. This result might not have been the case if we were using more extreme scenarios or received the portfolio closer to maturity. I have done some simulations when receiving the portfolio with both longer and shorter time to maturity with very similar results. Therefore, my conclusion is that no matter when we receive the portfolio, without reaching maturity, there will not be a significant change in the result.

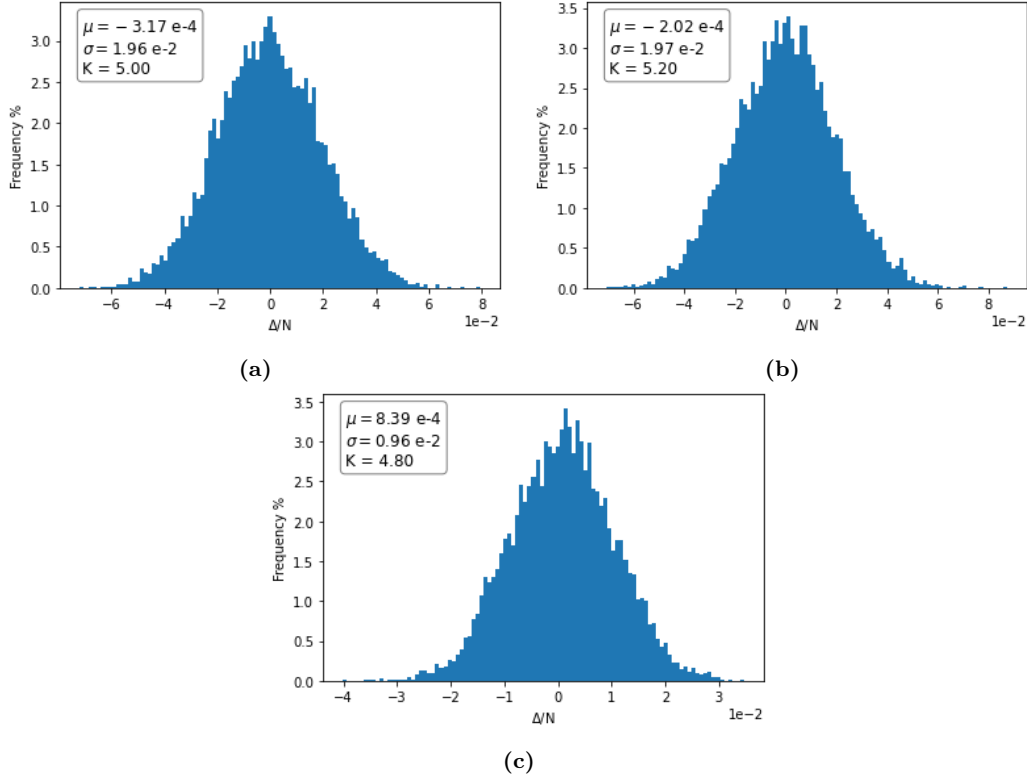


Figure 9: Results of hedging case 1. The delta exposure of the last day of hedging for each of the 10,000 outcomes when varying the strike rate K . The x-axis is normalized with respect to notional amount. In (a) $K = 5\%$, i.e., starting ATM, in (b) $K = 5.2\%$, OTM, and in (c) $K = 4.8\%$, ITM.

I do however believe that with more extreme scenarios we would see very different results. If we continued to use delta-hedging, our strategy would not differ one bit. There would not be a problem to find swaps with suitable deltas either. What would be the difference then? The difference would be that we are more exposed to randomness. This is because the exposure we have are almost entirely connected to the change in delta on the day of the auction. With delta-hedging we can't do anything to control this. The strategy is working very well, finding swaps that achieves almost exact delta neutrality each day. With this said, one could even argue that the first nine days we are holding the portfolio are unnecessary since it's only the last day that counts. I would argue that for stable market conditions, the first nine days are necessary because it provides a stability. And because we are using compression, we still only have one forward swap at the end of the hedging interval. In order to reduce our exposure to the one-day delay I think we would need to apply delta-gamma-hedging. With the information that the gamma would provide, we would be able to reduce the exposure of changes of the delta. The

reason why gamma is not in our initial scope is because it might be insignificant. When performing delta-gamma-hedging, the gamma is ignored if the value is too small. And under stable market conditions, which we have, I believe that this would be the case for most scenarios.

4.2 Case 2

In case 2, we receive the portfolio after 715 days, leaving us with five days left to maturity. This means that we will hedge with forward swaps for five days and thereafter use IRSs for the remaining five days. As for case 1, the last day of hedging was saved and in Fig(10) we can see the distributions from the simulations. We can interpret the result in the same way as in case 1.

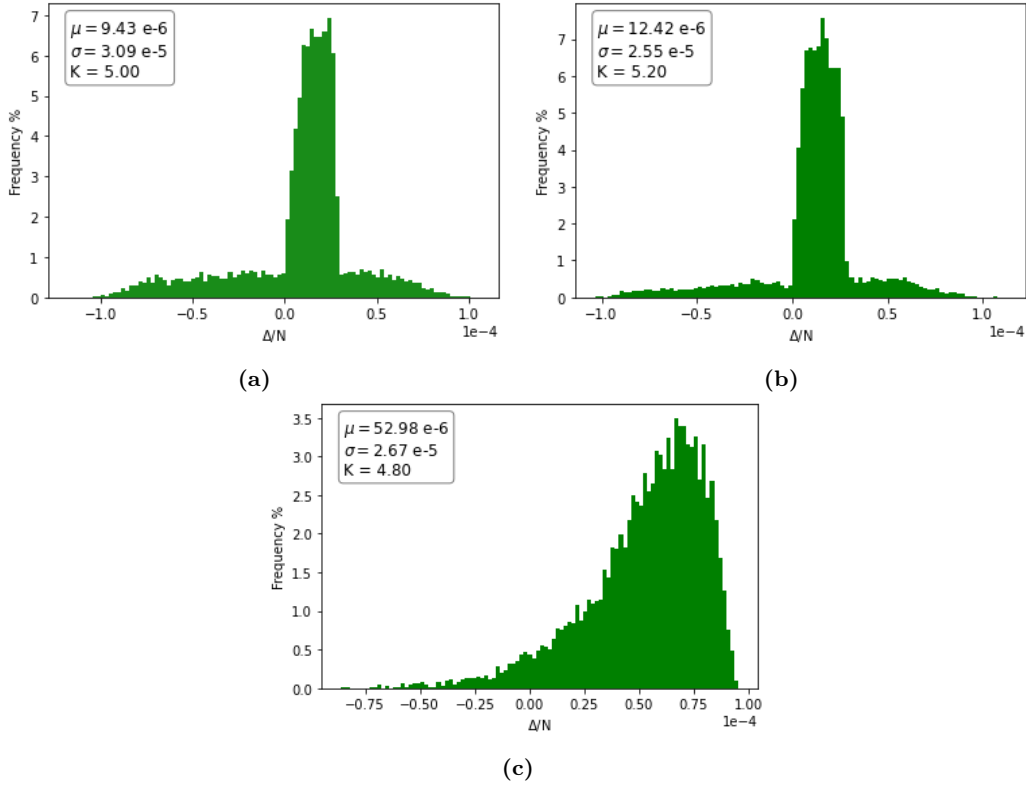


Figure 10: Results of hedging case 2. The delta exposure of the last day of hedging for each of the 10,000 outcomes when varying the strike rate K . The x-axis is normalized with respect to notional amount. In (a) $K = 5\%$, i.e., starting ATM, in (b) $K = 5.2\%$, OTM, and in (c) $K = 4.8\%$, ITM.

For this case we can clearly see a difference in the results. By visual inspection we

see that starting ATM and OTM results in a very similar distribution. However, when starting OTM we can see that there is a slightly lower σ with a result more centered around the mean, μ . The big difference is when we start ITM. We can see that μ is significantly higher and that the distribution is very different compared to the other two. Why we are seeing this distribution can be for several reasons. Before going into that lets go through what happens more thoroughly. After five days we have purchased five forward swaps to hedge the swaption. When maturity is reached, we will either have one IRS that comes from the swaption plus the five IRSs that comes from the forward swaps or only the five IRSs from the forward swaps. The IRS we receive from the swaption will have the default notional amount which is much higher than the adjusted notional we have created when purchasing the forward swaps. With a higher notional amount, we will have a bigger exposure to changes in the rates. Similarly, we will have a smaller exposure with smaller notional. However, the total delta should be somewhat close to being neutral if the swaption is exercised since that is what the forward swaps were purchased to counter. If the swaption is not exercised, the combined delta from the forward swaps would be far from delta neutral since the delta they were purchased to counter does not exist anymore. If we end up OTM when maturity is reached, the first day of hedging would therefore be with a notional that would be close to the one we had for the swaption in order to counter the combined delta from the forward swaps.

For the scenarios when we start ATM, we should theoretically have a somewhat 50/50 split if the swaption is exercised or not. For ITM/OTM the majority should end up exercised/not exercised. If we base our discussion from the point of view that the scenarios end up as just mentioned, why are we seeing a difference between ATM/OTM and ITM? If we start by comparing ATM and OTM we see that they are very similar but the spread, σ , is lower for OTM and μ is slightly higher. I think this is because when we begin ATM, we will have close to 50/50 split and since the results from OTM/ITM differ a lot we see a higher spread than in both other cases. However, I think the assumption that most cases did end up OTM is reasonable since ATM and OTM have such similar distributions. I don't really know why the results are slightly skewed to above zero.

When we start ITM, the result differs a lot from OTM/ATM and I'm not sure why that is. It might have something to do with that the IRS that comes from the swaption have some large changes which we miss with the one-day delay. As previously mentioned, it could also be a random effect or an error in the code. No matter what if there is an error or the results are skewed due to randomness there are still very good results. In Fig(10) we see that μ is of the magnitude 10^{-6} which is very small. Therefore, in this case the usage of delta-gamma-hedging, under stable market conditions, would be unnecessary since the numbers are even smaller than in case 1. As in case 1, more extreme scenarios would be interesting to investigate.

5 Conclusion and Future studies

With the increasing popularity of IRSs and the stricter regulations in Japan, and perhaps other countries in the near future, clearing of swaptions could be a good opportunity to get a bigger piece of the swap market. And to be able to offer a market for swaps and its derivatives the most important aspect, from a risk perspective, is to understand what price models to use. Since the black76 model is established as the "go to model" when pricing swaptions there wasn't any doubt that the model would work. But either way I felt the need to test the model with Monte Carlo. Another aspect to test the model with Monte Carlo is because the rate simulations are done with Monte Carlo. As we saw in Fig(7) the model seems accurate when testing toward a higher number of iterations.

As we know from the section about clearinghouses, they don't want to be a participant and they don't want any exposure to risks. Therefore, the aspect of being able to hedge a swaption is also of great importance. The result looks very promising for both cases, as we have seen in Fig(9) & Fig(10). There are however some assumptions we have done that I feel perhaps don't reflect the reality and would be interesting to investigate further. The first thing is the assumption that there are always suitable trades available on the market. In most cases there are always suitable trades available, but in our case when we adjust the notional, I think the trades can be hard to find. The amounts of the adjusted notional can be less than 1, which for me feels unreasonable. We have however had conversations with a contact working in Japan that confirmed that our strategy is in line with how they operate. He confirmed that you can't buy half a swap, and therefore they adjust the notional to find a suitable delta. He also told us that when they want to hedge, they put the responsibility to find the instrument they need on other clearing members that have better knowledge of the market. However, he has never given us any limits on how low/high a notional can be and therefore I am a bit concerned about that part. Since this thesis has a limited amount of time, we are forced to make certain assumptions and accept that they might not reflect the real world perfectly.

The other assumption we have done is a fixed discount rate and a fixed volatility throughout the simulations. The numbers I have chosen are not unrealistic, but perhaps it would be more accurate to recalculate the parameters with each rate update. The volatility could have been calculated with the market prices if we had used real data for the simulations. Realistically the discount rate would be updated with the rate changes and from real data the curves would be available. But as mentioned before, the thesis has a limited amount of time and limitations are necessary. Estimation of different volatilities and discount curves could be a thesis on its own.

For case 1 we used compression to minimize the number of swaps needed, could we do something similar for case 2? Our contact in Japan told us that they are using sensitivity buckets to reduce the number of IRSs needed. Sensitivity buckets can be thought of as the name intends, a bucket of IRSs. For each type of XIBOR the delta of the IRSs would be grouped up together based on their maturity. Meaning that all IRSs with maturity between 0-1, 1-2, 2-5 years, etc. would in this case be hedged with a single IRS of maturity 1,2 or 5 years. This would significantly reduce the number of IRSs needed

and would be a great substitution to compression.

As mentioned in the discussion about case 1&2, the scenarios would perhaps be the most interesting to investigate further. The best way to investigate this would be to use stress tests. Unfortunately, there was not enough time to start that part since it's a large area to implement. But if there would have been time, or if the chance to go back to this topic in the future there are events, we could use to perform stress tests. For example, you could use the data from the war between Russia and Ukraine in 2022. During the war the Russian interest rate rose from around 9 to 20%. This is incredibly extreme and before that event I think that such an increase would be too extreme to use in a stress test. But now that it has occurred once, it should be tested as an extreme but plausible scenario. We can draw a parallel to 2008, my guess is that before that crisis, such extreme events seemed not plausible but these days that data is used in stress tests.

In conclusion I think it's fair to say that the model works, and our hedging strategy performs well, even if it's only for stable market conditions. We can also conclude that the usage of compression would simplify the hedging process significantly since a defaulted portfolio can consist of hundreds or perhaps thousands of swaptions. Even if the results are good there are several factors, as mentioned above, that could increase the performance of our current model. These factors could also broaden the usage area to also include extreme market conditions.

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